Nickolas-Adrian Papamichail

SCIENTIFIC SOFTWARE ENGINEERING | C++ | COMPUTATIONAL MODELING

Athens, Greece | papamichailnickolas@gmail.com | github.com/IanPapamichail

LinkedIn: Nickolas-Adrian Papamichail | English: professional working proficiency

PROFILE

Informatics and Telecommunications undergraduate focused on C++ software and computational research. Experience with numerical modeling, Python scientific computing and algorithm evaluation; developing a C++ simulation architecture for BioMapX. Seeking junior scientific software, simulation or HPC engineering roles.

TECHNICAL SKILLS

Programming C++, C, Python
Scientific computing NumPy, SciPy, numerical modeling, linear algebra, probability
Development Git, Linux, CMake, VS Code; SSH workflows
Research methods simulation design, benchmark analysis, reproducible experiments

SELECTED PROJECTS

BioMapX / Q-Chaperone | C++ scientific simulation architecture · in development

- Defined configurable simulation inputs including temperature, pressure, pH, time step and random seed for molecular modeling workflows.
- Designed a modular backend to separate simulation state, computational workloads and future hybrid HPC scheduling.

Cognitive Optimizer | Python, Qiskit · bachelor thesis experiment

- Implemented and evaluated a circuit transformation for merging phase rotations; checked consistency with memory enabled and disabled.
- Benchmarked 12-qubit VQE and 10-qubit synthetic circuits, recording gate count, depth and two-qubit gate count.

Graphene Lattice Simulation | Hamiltonian modeling · numerical analysis

- Modeled a graphene lattice using a Hamiltonian formulation and analyzed eigenvalues to investigate electronic states.

EDUCATION

University of Ioannina | BSc in Informatics and Telecommunications
GPA 8.75/10 | Expected graduation September 2026
Relevant foundations Algorithms, linear algebra, probability and scientific computing

SELECTED TRAINING

Elements of Supercomputing | Kajaani University of Applied Sciences, 2026
Integrating Quantum and High-Performance Computing | IBM, 2026